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**Superconductivity-Induced Shift in Resonant Field in a
Superconductor-Ferromagnet-Superconductor Trilayer**

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Chapter 1

Introduction

Descrizione generale del lavoro. La farò come ultima cosa: dare una descrizione generale di una cosa che esiste già è molto più facile che dare una descrizione generale di una cosa che non esiste ancora. [4] [3] [2] [5] [6] [1]

Ferromagnetism and superconductivity were long thought to be incompatible due to their opposite orderings. Ferromagnetism is linked to the exchange interaction that leads to symmetric spin states, therefore generally a net magnetisation. Superconductivity is mediated by Cooper pairs (CPs) which are comprised of two electrons with equal and opposite momenta, bound together through lattice interaction, phonons. This thesis presents an investigation on the magnetization dynamics in superconductor-ferromagnet-superconductor thin films multilayers layered films conducted during my summer internship at the Cavendish Laboratory in Cambridge. After a theoretical introduction on ferromagnetism, ferromagnetic resonance superconductivity and the interface effect in superconductor-ferromagnet thin films interfaces, this thesis presents a. Superconductivity and ferromagnetism were long

Chapter 2

Theoretical Background

2.1 Ferromagnetism

2.1.1 Maxwell's equations in matter

The time evolution of the electric field $\mathbf{E}$ and the magnetic flux density $\mathbf{B}$ is governed by Maxwell's equations:

$$\left\{ \begin{array}{l} \nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0} \end{array} \right. \quad (2.1)$$

$$\left\{ \begin{array}{l} \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \end{array} \right. \quad (2.2)$$

$$\left\{ \begin{array}{l} \nabla \cdot \mathbf{B} = 0 \end{array} \right. \quad (2.3)$$

$$\left\{ \begin{array}{l} \nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t} \end{array} \right. \quad (2.4)$$

where ρ is the electric charge density, $\mathbf{J}$ is the electric current density and the values of the constants ε_0 and μ_0 in SI units are:

$$\varepsilon_0 = 8.8541878188 \frac{\text{C}}{\text{Nm}^2}$$
$$\mu_0 = 1.25663706127 \frac{\text{N}}{\text{A}^2}$$

These equations, together with the Lorentz force law,

$$\mathbf{F} = q (\mathbf{E} + \mathbf{v} \times \mathbf{B}) \quad (2.5)$$

which gives the force acting on a particle of charge q moving with velocity $\mathbf{v}$ in an electromagnetic field $(\mathbf{E}, \mathbf{B})$, constitute the theory of classical electrodynamics.

From 2.3 we can write $\mathbf{B}$ as the curl of a vector field $\mathbf{A}$ called the vector potential

$$\mathbf{B} = \nabla \times \mathbf{A} \quad (2.6)$$

substituting 2.6 in 2.2 we get

$$\nabla \times \left(\mathbf{E} + \frac{\partial \mathbf{A}}{\partial t} \right) = \mathbf{0} \quad (2.7)$$

which implies that $\mathbf{E} + \frac{\partial \mathbf{A}}{\partial t}$ can be written as the negative gradient of a scalar field V , called the scalar potential

$$\begin{aligned} \mathbf{E} + \frac{\partial \mathbf{A}}{\partial t} &= -\nabla V \\ \mathbf{E} &= -\nabla V - \frac{\partial \mathbf{A}}{\partial t} \end{aligned} \quad (2.8)$$

The energy density of the electromagnetic field is given by

$$u = \frac{\varepsilon_0}{2} E^2 + \frac{1}{2\mu_0} B^2. \quad (2.9)$$

Therefore, the total energy required to generate an electromagnetic field $(\mathbf{E}, \mathbf{B})$ from a configuration with no field is

$$W = \int u d^3\mathbf{r} = \int \left(\frac{\varepsilon_0}{2} E^2 + \frac{1}{2\mu_0} B^2 \right) d^3\mathbf{r} \quad (2.10)$$

When dealing with matter, it is useful to introduce the following macroscopic quantities:

the polarization vector, defined as the electric dipole moment per unit volume:

$$\mathbf{P} = \frac{d\mathbf{p}}{dV} \quad (2.11)$$

the magnetization vector, defined as the magnetic dipole moment per unit volume:

$$\mathbf{M} = \frac{d\mathbf{m}}{dV} \quad (2.12)$$

the electric displacement field:

$$\mathbf{D} = \varepsilon_0 \mathbf{E} + \mathbf{P} \quad (2.13)$$

the magnetic field:

$$\mathbf{H} = \frac{\mathbf{B}}{\mu_0} - \mathbf{M} \quad (2.14)$$

In the presence of polarization and magnetization, the charge and current densities can be expressed as

$$\rho = \rho_f + \rho_b \quad (2.15)$$

where $\rho_b = \nabla \cdot \mathbf{P}$ is the bound charge density due to polarization and ρ_f is the free charge density due to macroscopic charges, and

$$\mathbf{J} = \mathbf{J}_f + \mathbf{J}_b + \mathbf{J}_p \quad (2.16)$$

where $\mathbf{J}_b = \nabla \times \mathbf{M}$ is the bound current density due to magnetization, $\mathbf{J}_p = \frac{\partial \mathbf{P}}{\partial t}$ is the polarization current, and $\mathbf{J}_f$ is the free current density due to macroscopic currents.

Using these definitions, Maxwell's equations can be rewritten in the equivalent form:

$$\left\{ \begin{array}{l} \nabla \cdot \mathbf{D} = \rho_f \\ \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \\ \nabla \cdot \mathbf{B} = 0 \\ \nabla \times \mathbf{H} = \mathbf{J}_f + \frac{\partial \mathbf{D}}{\partial t} \end{array} \right. \quad (2.17)$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \quad (2.18)$$

$$\nabla \cdot \mathbf{B} = 0 \quad (2.19)$$

$$\nabla \times \mathbf{H} = \mathbf{J}_f + \frac{\partial \mathbf{D}}{\partial t} \quad (2.20)$$

The electric and magnetic susceptibilities are defined as:

$$\chi_e = \frac{1}{\varepsilon_0} \frac{\partial \mathbf{P}}{\partial \mathbf{E}} \quad (2.21)$$

$$\chi_m = \frac{\partial \mathbf{M}}{\partial \mathbf{H}} \quad (2.22)$$

Since $\frac{\partial \mathbf{A}}{\partial \mathbf{B}}$ is a tensor such that

$$\left(\frac{\partial \mathbf{A}}{\partial \mathbf{B}} \right)_{ij} = \frac{\partial A_i}{\partial B_j},$$

χ_e and χ_m are in general tensors depending on position $\mathbf{r}$. If a material is homogeneous, they do not depend on $\mathbf{r}$; if it is isotropic, they reduce to scalar quantities. Materials with $\chi_m > 0$ are called paramagnets, while those with $\chi_m < 0$ are called diamagnets.

2.1.2 Properties of ferromagnetic materials

2.1.3 Energy of ferromagnetic materials

The interplay between multiple interactions determines the domain structure in ferromagnets.

2.1.4 Ferromagnetic resonance

A system with angular momentum $\mathbf{L}$ and magnetic moment $\boldsymbol{\mu} = \gamma\mathbf{L}$ subject to an external magnetic field $\mathbf{H}$ evolves according to the following equation

$$\frac{d\mathbf{L}}{dt} = \boldsymbol{\mu} \times \mathbf{H} \quad (2.23)$$

The magnetization of a ferromagnetic sample is proportional to the sum of the magnetic moments of its electrons. Its dynamics is therefore expected to follow the equation

$$\frac{d\mathbf{M}}{dt} = \gamma\mathbf{M} \times \mathbf{H} \quad (2.24)$$

where $\mathbf{H}$ is the external applied magnetic field and γ is the gyromagnetic ratio of an electron. However this form predicts perpetual precession without relaxation while experimentally, the magnetization is observed to align with the applied field after a finite time. To account for this, damping is introduced through the Gilbert term, which models a viscous-like relaxation. The full dynamics is described by the Landau–Lifshitz–Gilbert (LLG) equation

$$\frac{1}{\gamma} \frac{d\mathbf{M}}{dt} = \mathbf{M} \times \mathbf{H} + \frac{\alpha}{M_s} \left(\mathbf{M} \times \frac{d\mathbf{M}}{dt} \right) \quad (2.25)$$

where α is the dimensionless Gilbert damping parameter and M_s is the saturation magnetization.

Ferromagnetic resonance (FMR) is observed when a ferromagnetic material is subjected to both a constant (DC) magnetic field and a perpendicular oscillating (AC) magnetic field, typically at microwave frequencies.

When an external magnetic field $\mathbf{H}$ is applied, the electronic energy levels split due to the Zeeman effect. Resonance occurs when the frequency of the transverse oscillating field matches the energy difference between Zeeman levels

$$\Delta E = g\mu_B B = h\nu \quad (2.26)$$

From the LLG equation, one can derive the relation between the resonance frequency ω_{res} and the resonance field H_{res} , known as the Kittel equation. Its form depends on the sample geometry and the orientation of the static field. For a thin film with an in-plane DC field, it reads

$$\omega_{res} = \gamma \sqrt{H_{res}(H_{res} + M_{eff})} \quad (2.27)$$

The Gilbert damping can also be determined by measuring the linewidth of the resonance peak as a function of frequency via the relation

$$\Delta H = \Delta H_0 + \frac{2\pi f \alpha}{\mu_0 \gamma} \quad (2.28)$$

2.2 Superconductivity

2.2.1 Conduction in metals

The conductivity of a material is defined by the constitutive relation:

$$\mathbf{J} = \sigma \mathbf{E} \quad (2.29)$$

Here $\mathbf{J}$ and $\mathbf{E}$ are, respectively, the electrical current density and the electric field at some point in a material. According to Drude's model, the conductivity of a metal is given by

$$\sigma = \frac{ne^2\tau}{m} \quad (2.30)$$

Where n is the electron density, e is the elementary charge, m is the mass of an electron, and τ is the average lifetime for free motion of the electrons between collisions with impurities, ions of the lattice, and other electrons. This means that the resistivity ρ , defined as $\rho = \sigma^{-1}$, is given by

$$\rho = \frac{m}{ne^2}\tau^{-1} \quad (2.31)$$

Since the scattering by impurities, electrons, and phonons is independent, the total scattering rate τ^{-1} can be written as

$$\tau^{-1} = \tau_{\text{imp}}^{-1} + \tau_{\text{ph}}^{-1} + \tau_{\text{el}}^{-1} \quad (2.32)$$

The impurity scattering rate is essentially independent of temperature, the electron scattering rate is proportional to T^2 (where T is the temperature), and at low temperatures (well below the Debye temperature) the phonon

scattering rate is proportional to T^5 . Thus, the resistivity of a material is given by

$$\rho = \rho_0 + aT^2 + bT^5 \quad (2.33)$$

Even though it is an extremely simplified model that does not take into account the effective mass and the fact that only the electrons at energies near the Fermi level are effectively involved in conduction, Drude's model successfully predicts the temperature dependence of resistivity for many metals. However there are some materials for which something different happens: when a superconducting material is cooled below a certain critical temperature T_c , its resistivity drops to zero. This is an example of a phase transition, which is a physical process where the properties of a macroscopic system change abruptly upon variation of a thermodynamic variable. Figure 2.1 shows the difference in the temperature dependence of resistivity between a superconductor and a normal metal.

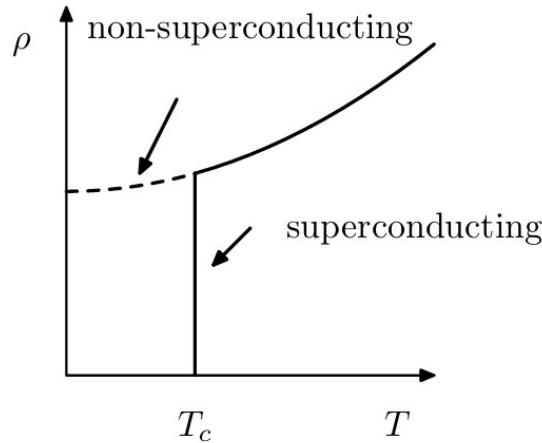


Figure 2.1: Temperature dependence of the resistivity for a normal metal and a superconducting metal.

2.2.2 The Meissner effect

Even though superconductors were historically discovered through the observation of zero resistivity, today this property is not regarded as the true definition of superconductivity. The fundamental proof that a material is superconducting is the demonstration of the Meissner effect, namely the complete expulsion of a weak external magnetic field. This is because the magnetic

behavior of superconductors below their critical temperature cannot be fully explained by zero resistivity alone.

As an example, let us consider a superconducting sample in the four configurations shown in Figure 2.2.

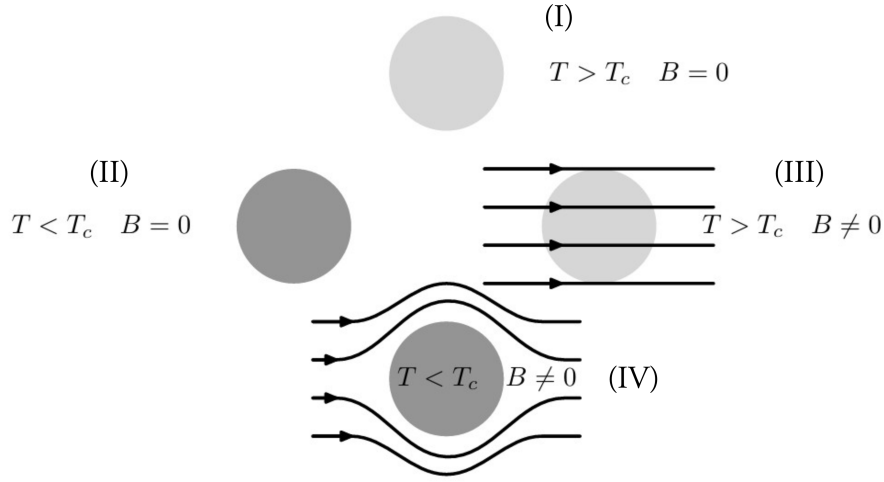


Figure 2.2

Suppose initially the sample is in its normal state, $T > T_c$, and the external magnetic field is zero, as in (I). If we cool the sample below T_c while keeping the field zero, we obtain configuration (II). Then, when the external field is gradually turned on, the field inside the sample must remain zero, as in (IV). This follows from the fact that with $\sigma = 0$ and a finite value of $\mathbf{J}$ in 2.29, we must have

$$\mathbf{E} = \mathbf{0} \quad (2.34)$$

at every point inside the superconducting sample. Combining this with Maxwell's equation 2.2, we obtain

$$\frac{\partial \mathbf{B}}{\partial t} = \mathbf{0} \quad (2.35)$$

throughout the superconductor. This implies that if the initial condition is $\mathbf{B} = \mathbf{0}$, it will remain so as long as the sample stays in the superconducting state, even when an external magnetic field is applied.

If instead we start from (I) and turn on the external magnetic field, the field will penetrate the sample as in (II). However, if we then cool the sample below T_c while maintaining the external field, we again obtain $\mathbf{B} = \mathbf{0}$ inside the superconductor, showing that the magnetic field is completely expelled. This behavior cannot be explained solely in terms of zero resistance. This is the difference between a superconductor and a perfect conductor: a perfect conductor is a material that preserves $\mathbf{B}$ no matter what its value is, a superconductor instead is a material that expels $\mathbf{B}$ so not only it cannot change but it has to be zero.

The complete expulsion of the magnetic field means that, for weak magnetic fields, superconductors behave as perfect diamagnets. In fact, since $\mathbf{B} = \mathbf{0}$, from 2.14 we have

$$\mathbf{M} = -\mathbf{H}, \quad (2.36)$$

and combining 2.36 with 2.22, we find

$$\chi = \frac{\partial \mathbf{M}}{\partial \mathbf{H}} = -1. \quad (2.37)$$

The best way to detect superconductivity in some unknown sample is therefore to measure its susceptibility. If the sample is fully superconducting then χ_m as a function of T will be something like the sketch in figure 2.3. Above T_c it is a constant normal state value which is usually small and positive (paramagnetic). Below T_c one will find $\chi_m = -1$, evidence for the Meissner effect. This is usually considered much more reliable evidence for superconductivity in a sample than zero resistance alone would be.

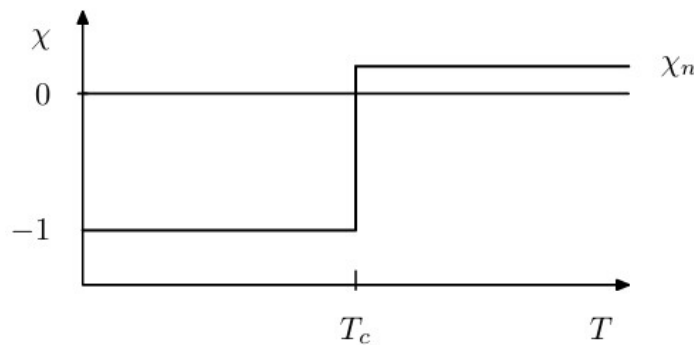


Figure 2.3: Magnetic susceptibility of a superconductor as a function of temperature.

2.2.3 Type I and type II superconductors

As the external field applied to a superconductor becomes stronger it turns out that either one of two possible things can happen.

For type I superconductors, the $\mathbf{B}$ field remains zero inside the material until suddenly the superconductivity is destroyed. The field where this happens is called the critical field, H_c . The magnetization of a type I obeys $\mathbf{M} = -\mathbf{H}$ for all fields less than H_c , and then becomes very close to zero for fields above H_c .

For type II superconductors there are two different critical fields, denoted H_{c1} , the lower critical field, and H_{c2} the upper critical field. For small values of applied field $\mathbf{H}$ the Meissner effect leads to $\mathbf{M} = -\mathbf{H}$ and there is no magnetic flux density inside the sample, $\mathbf{B} = \mathbf{0}$. However, once the field exceeds H_{c1} , magnetic flux does start to enter the superconductor and hence $\mathbf{B} \neq \mathbf{0}$, and $\mathbf{M}$ is closer to zero than the full Meissner value of $-\mathbf{H}$. Upon increasing the field H further the magnetic flux density gradually increases, until finally at H_{c2} the superconductivity is destroyed and $\mathbf{M} \approx \mathbf{0}$. The phase below H_{c1} is normally denoted the Meissner state, while the phase between H_{c1} and H_{c2} is the Abrikosov or mixed state.

The behaviour of the magnetization for type I and type II superconductors is sketched in figure 2.4.

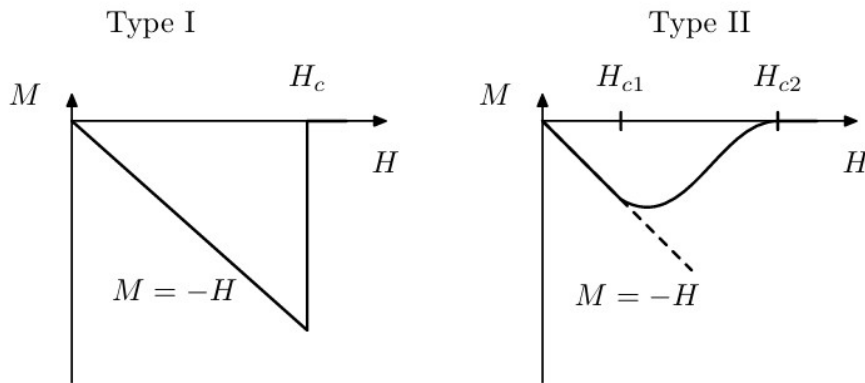


Figure 2.4: The magnetization as a function of the magnetic field in type I and type II superconductors.

As a function of the temperature the critical fields vary, and they all approach zero at the critical temperature T_c . The typical phase diagrams of type I and type II superconductors, as a function of H and T are shown in figure.

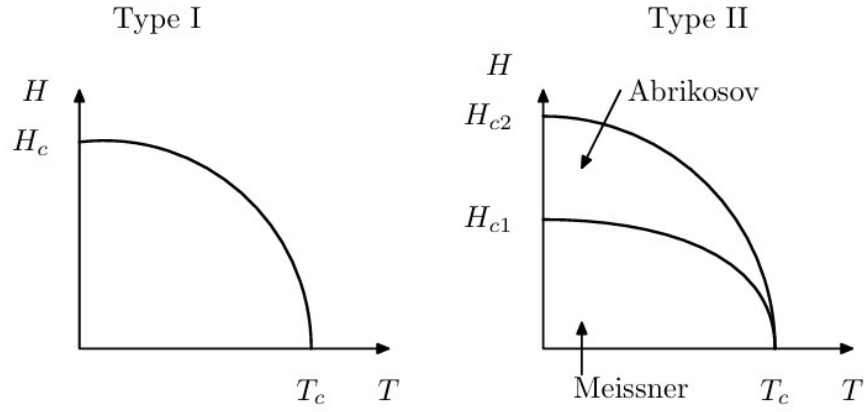


Figure 2.5: The H - T phase diagram of type I and type II superconductors.

The physical reason for the thermodynamic phase between H_{c1} and H_{c2} is the presence of vortices: the magnetic field can enter the superconductor in the form of vortices, as shown in figure 2.6. Each vortex consists of a region of circulating supercurrent around a small central core which has essentially become normal metal. The magnetic field is able to pass through the sample inside the vortex cores, and the circulating currents serve to screen out the magnetic field from the rest of the superconductor outside the vortex. This allows the bulk of the material to remain superconducting, while also allowing a finite average magnetic flux density $\mathbf{B}$ to pass through.

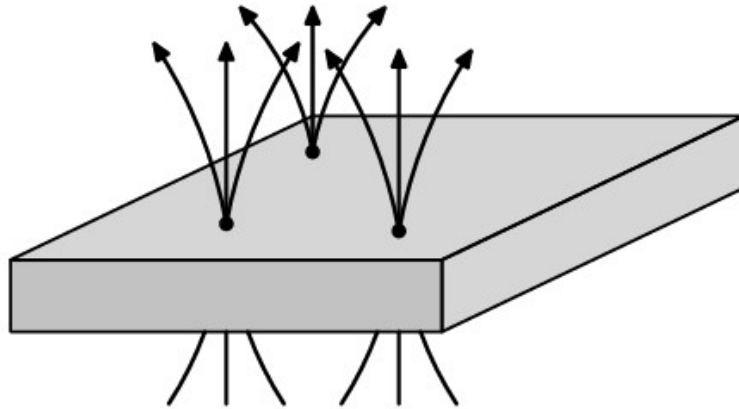


Figure 2.6: Vortices in a type II superconductor.

2.2.4 The London Equations

The behaviour of the current density $\mathbf{J}$ in a superconducting material is described by the following equations:

$$\frac{\partial \mathbf{J}}{\partial t} = \frac{\mathbf{E}}{\mu_0 \lambda^2} \quad (2.38)$$

$$\nabla \times \mathbf{J} = -\frac{\mathbf{B}}{\mu_0 \lambda^2} \quad (2.39)$$

They are called London equations were the first successful attempt at characterizing the electrodynamic behavior of superconductors. Here λ is called the London penetration depth and is a measure of how deep a magnetic field is allowed to penetrate into the superconductor.

$$\mathbf{J} = -\frac{\mathbf{A}}{\mu_0 \lambda^2} \quad (2.40)$$

London equations are the most precise way to define a superconducting material and any model of superconductivity must be able to predict this equations.

London equations can also be used to explain the Meissner effect. By substituting Maxwell's equation into equation we get

$$\nabla \times (\nabla \times \mathbf{B}) = -\frac{\mathbf{B}}{\lambda^2} \quad (2.41)$$

using the vector identity

$$\nabla \times (\nabla \times \mathbf{A}) = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A} \quad (2.42)$$

and using Maxwell's equation we get

$$\nabla^2 \mathbf{B} = \frac{\mathbf{B}}{\lambda^2} \quad (2.43)$$

By solving this equation we can see that in the pure superconducting state the only field allowed is exponentially damped as we go in from an external surface. If we consider for example a semi-infinite superconductor occupy the space on the positive side of the x axis with a uniform field as in figure.

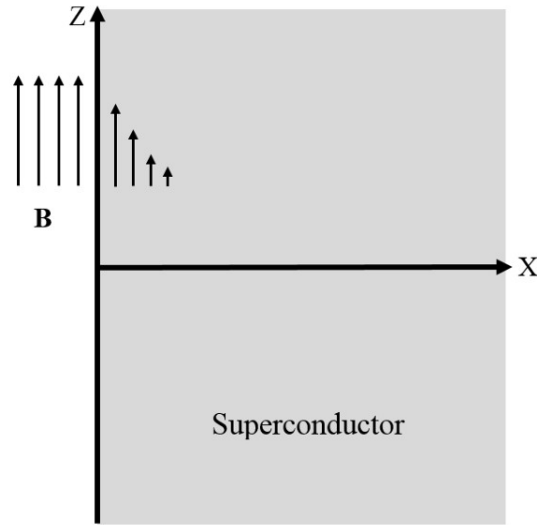


Figure 2.7

$$\frac{\partial^2 B_z}{\partial x^2} = \frac{B_z}{\lambda^2} \quad (2.44)$$

the solution to this ODE is

$$B_z(x) = Ae^{-\frac{x}{\lambda}} + Be^{\frac{x}{\lambda}} \quad (2.45)$$

since a field with $B \neq 0$ would require an infinite amount of energy (as we can see from 2.10), we must have $B = 0$, combining this information with the boundary condition $B_z(x = 0) = B_0$, we get

$$B_z(x) = B_0 e^{-\frac{x}{\lambda}} \quad (2.46)$$

this is illustrated in figure 2.8.

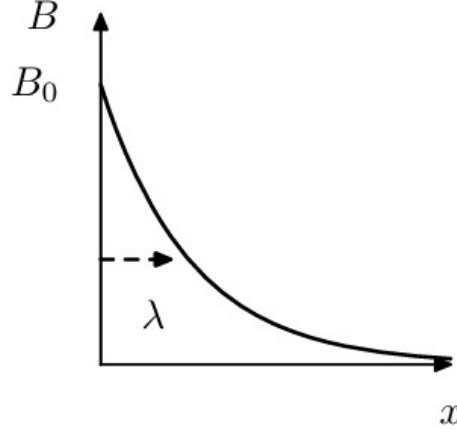


Figure 2.8: Exponential decay of the magnetic field near the surface of a superconductor in the Meissner state.

The London equation can be derived with a simple model where a fraction of the electrons are treated as a superfluid and move around the material without encountering any resistance. Therefore the equation of motion for any of these superconducting electrons is

$$m \frac{d\mathbf{v}}{dt} = -e\mathbf{E} \quad (2.47)$$

the current density can be written as

$$\mathbf{J} = -en_s\mathbf{v} \quad (2.48)$$

therefore

$$\frac{\partial \mathbf{J}}{\partial t} = -en_s \frac{\partial \mathbf{v}}{\partial t} = \frac{e^2 n_s}{m} \frac{\partial \mathbf{E}}{\partial t} \quad (2.49)$$

which is the first London equation. Confronting with we get

2.2.5 The Ginzburg-Landau model

The Ginzburg-Landau model is a phenomenological theory that describes the superconductive phase transition from a thermodynamical perspective and it can be shown that it follows from the BCS theory. When describing a phase transition an order parameter, is a variable that is zero when the

transition has not happened and is different from zero when the transition has happened, an example of this is the spontaneous magnetization of a ferromagnetic material below its Curie temperature. The assumption of the Ginzburg-Landau theory is that a superconductive material is described by a complex order parameter ψ such that

$$\psi = \begin{cases} 0 & T > T_c \\ \psi(T) \neq 0 & T < T_c \end{cases} \quad (2.50)$$

Further more, it assumes that the Helmholtz free energy of the superconductor (defined as $F = U - TS$) depends smoothly on the parameter ψ . Since ψ is complex and the free energy must be real, the energy can only depend on $|\psi|$. Furthermore, since ψ goes to zero at the critical temperature, T_c , we can Taylor expand the free energy in powers of $|\psi|$. For temperatures close to T_c only the first two terms in the expansion should be necessary, and so the free energy density ($f = F/V$) must be of the form:

$$f_s(T) \approx f_n(T) + a(T)|\psi|^2 + \frac{1}{2}b(T)|\psi|^4 \quad (2.51)$$

Here $f_s(T)$ and $f_n(T)$ are the superconducting state and normal state free energy densities, respectively. Clearly equation 2.51 is the only possible function which is real for any complex ψ near $\psi = 0$ and which is a differentiable function of ψ and ψ^* near to $\psi = 0$. The parameters $a(T)$ and $b(T)$ are, in general, temperature dependent phenomenological parameters of the theory. However it is assumed that they must be smooth functions of temperature. We must also assume that $b(T)$ is positive, since otherwise the free energy density would have no minimum, which would be unphysical (or we would have to extend the expansion to include higher powers such as $|\psi|^6$). Plotting $f_s - f_n$ as a function of $|\psi|$ is easy to see that there are two possible curves, depending on the sign of the parameter $a(T)$, as shown in figure. In the case $a(T) < 0$, the curve has one minimum at $\psi = 0$. On the other hand, for $a(T) > 0$ there are minima wherever $|\psi|^2 = -\frac{a(T)}{b(T)}$. The model assumes that at high temperatures, above T_c , we have $a(T)$ positive, and hence the minimum free energy solution is one with $\psi = 0$, namely the normal state. But if $a(T)$ gradually decreases as the temperature T is reduced, then the state of the system will change suddenly when we reach the point $a(T) = 0$. Below this temperature the minimum free energy solution changes to one with $\psi \neq 0$. Therefore we can identify the temperature where $a(T)$ becomes zero as the critical temperature T_c .

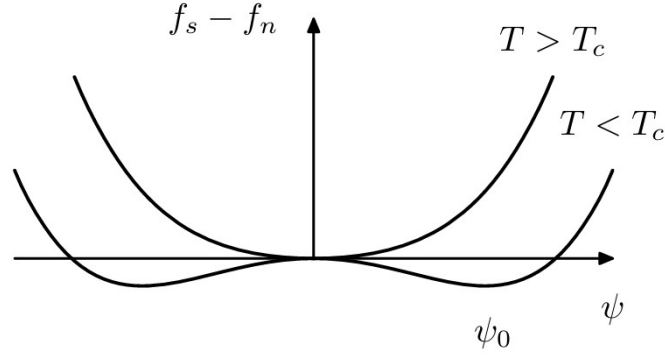


Figure 2.9: Free energy difference between the normal and superconducting states (per unit volume) as a function of the order parameter ψ . For $T < T_c$ the free energy has a minimum at ψ_0 , while for $T \geq T_c$ the only minimum is at $\psi = 0$.

Near to this critical temperature, T_c , assuming that the coefficients $a(T)$ and $b(T)$ change smoothly with temperature, we can make a Taylor expansion

$$\begin{aligned} a(T) &\approx \dot{a}(T - T_c) \\ b(T) &\approx b \end{aligned} \quad (2.52)$$

where $\dot{a}$ and b are two phenomenological constants. Then for temperatures just above T_c , $a(T)$ will be positive, and we have the free energy minimum, $\psi = 0$. On the other hand, just below T_c we will have minimum energy solutions with non-zero $|\psi| \neq 0$, as seen in figure. In terms of the parameters a and b it is easy to see that

$$|\psi| = \begin{cases} 0 & T > T_c \\ \sqrt{\frac{\dot{a}}{b}} \sqrt{T - T_c} & T < T_c \end{cases} \quad (2.53)$$

The corresponding curve of $|\psi|$ as a function of temperature T is shown in figure.

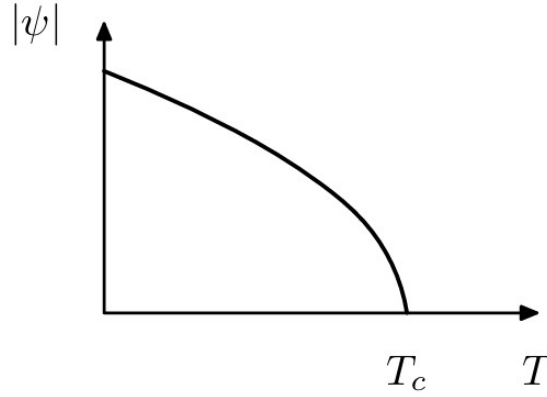


Figure 2.10: Order parameter magnitude as a function of temperature in the Ginzburg-Landau model.

The complete Ginzburg-Landau theory of superconductivity also allows for the possibility that the order parameter depends on position $\psi(\mathbf{r})$. The theory postulates that the free energy is as given above, together with a new term depending on the gradient of $\psi(\mathbf{r})$. With this term free energy density becomes

$$f_s(T) = f_n(T) + \frac{\hbar^2}{2m^*} |\nabla \psi(\mathbf{r})|^2 + a(T) |\psi(\mathbf{r})|^2 + \frac{b(T)}{2} |\psi(\mathbf{r})|^4 \quad (2.54)$$

Setting $\psi(\mathbf{r})$ to a constant value, we see that the parameters $a(T)$ and $b(T)$ are the same as for the bulk theory described in the previous section. The new parameter m^* determines the energy cost associated with gradients in $\psi(\mathbf{r})$. It has dimensions of mass, and it plays the role of an effective mass for the quantum system with macroscopic wave function $\psi(\mathbf{r})$.

In order to find the order parameter at the equilibrium we must minimize the total free energy of the system

$$F_s(T) = F_n(T) + \int \left(\frac{\hbar^2}{2m^*} |\nabla \psi(\mathbf{r})|^2 + a(T) |\psi(\mathbf{r})|^2 + \frac{b(T)}{2} |\psi(\mathbf{r})|^4 \right) d^3\mathbf{r} \quad (2.55)$$

To find the minimum, we must evaluate the infinitesimal change in the total free energy δF and set it equal to zero for any change in the order parameter

$\delta\psi$

$$\begin{aligned}
\delta F_s &= \delta \int \left(\frac{\hbar^2}{2m^*} |\nabla\psi|^2 + a|\psi|^2 + \frac{b}{2} |\psi|^4 \right) d^3\mathbf{r} \\
&= \int \delta \left(\frac{\hbar^2}{2m^*} \nabla\psi \cdot \nabla\psi^* + a\psi\psi^* + \frac{b}{2} \psi^2(\psi^*)^2 \right) d^3\mathbf{r} \\
&= \int \left(\frac{\hbar^2}{2m^*} (\nabla\delta\psi \cdot \nabla\psi^* + \nabla\psi \cdot \nabla\delta\psi^*) \right. \\
&\quad \left. + a(\psi\delta\psi^* + \psi^*\delta\psi) + b|\psi|^2(\psi\delta\psi^* + \psi^*\delta\psi) \right) d^3\mathbf{r} \tag{2.56} \\
&= \int \left(\frac{\hbar^2}{2m^*} \nabla\delta\psi^* \cdot \nabla\psi + \delta\psi^*(a\psi + b\psi|\psi|^2) \right) d^3\mathbf{r} \\
&\quad + \int \left(\frac{\hbar^2}{2m^*} \nabla\delta\psi \cdot \nabla\psi^* + \delta\psi(a\psi^* + b\psi^*|\psi|^2) \right) d^3\mathbf{r}
\end{aligned}$$

The two terms involving gradients can be integrated by parts to obtain

$$\begin{aligned}
\delta F_s &= \int \delta\psi^* \left(-\frac{\hbar^2}{2m^*} \nabla^2\psi + a\psi + b\psi|\psi|^2 \right) d^3\mathbf{r} \\
&\quad + \int \delta\psi \left(-\frac{\hbar^2}{2m^*} \nabla^2\psi^* + a\psi^* + b\psi^*|\psi|^2 \right)^* d^3\mathbf{r} \tag{2.57}
\end{aligned}$$

since this must hold for an arbitrary variation $\delta\psi$, we must have

$$-\frac{\hbar^2}{2m^*} \nabla^2\psi + a\psi + b\psi|\psi|^2 = 0 \tag{2.58}$$

This is a Schrödinger-like equation that allows us to find the order parameter $\psi(\mathbf{r})$ in equilibrium conditions. However, unlike the usual Schrödinger equation, this is a non-linear equation because of the second term in the bracket and because of this non-linearity the principle of superposition does not apply.

The effective non-linear Schrödinger equation 2.58, has several useful applications. In particular, it can be used to study the response of the superconducting order parameter to external perturbations. Consider a simple model for the interface between a normal metal and a superconductor. Suppose that the interface lies in the yz plane separating the normal metal in the $x < 0$ region from the superconductor in the $x > 0$ region. On the normal metal side of the interface the superconducting order parameter, $\psi(\mathbf{r})$, must be zero. Assuming that $\psi(r)$ must be continuous, we must therefore solve

the non-linear Schrödinger equation

$$-\frac{\hbar^2}{2m^*} \frac{d^2\psi}{dx^2} + a\psi + b\psi|\psi|^2 = 0 \quad (2.59)$$

in the region $x > 0$ with the boundary condition $\psi(0) = 0$. This equation can be solved directly and its solution is

$$\psi(x) = \psi_0 \tanh\left(\frac{x}{\sqrt{2}\xi(T)}\right) \quad (2.60)$$

as shown in figure 2.11.

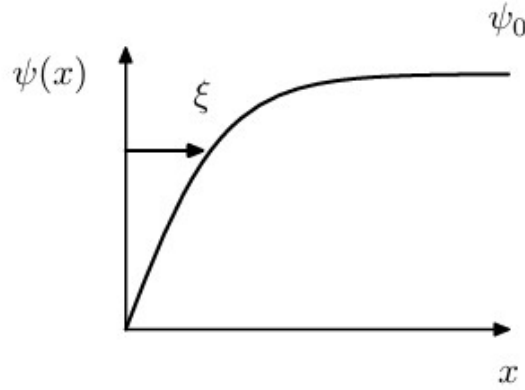


Figure 2.11: Order parameter of a superconductor near a surface. It recovers to its bulk value ψ_0 over a length scale of the coherence length, ξ .

Here ψ_0 is the value of the order parameter in the bulk far from the surface and the parameter $\xi(T)$ is defined by

$$\xi(T) = \frac{\hbar}{\sqrt{2m^*|a(T)|}} \quad (2.61)$$

This quantity, which has dimensions of length, is called the Ginzburg-Landau coherence length. It is an important physical parameter characterizing the superconductor. In figure 2.11 one can see that $\xi(T)$ is a measure of the distance from the surface over which the order parameter has recovered back to nearly its bulk value. The coherence length arises in almost all problems of inhomogeneous superconductors. Using $a(T) = a(T - T_c)$ the coherence length can be rewritten as

$$\xi(T) = \xi(0)|t|^{-\frac{1}{2}} \quad (2.62)$$

where

$$t = \frac{T - T_c}{T_c} \quad (2.63)$$

is called the reduced temperature. This expression makes it clear that the coherence length diverges at the critical temperature.

In the presence of a magnetic field the expression 2.54 must be extended according to the usual substitution

$$-i\hbar\nabla \rightarrow -i\hbar\nabla - q\mathbf{A} \quad (2.64)$$

For all known superconductors it turns out that the appropriate charge q is $-2e$ and the reason for this becomes clear in the BCS theory. With this replacement the Ginzburg-Landau free energy density of the superconductor becomes

$$f_s(T) = f_n(T) + \frac{1}{2m^*} |(-i\hbar\nabla + 2e\mathbf{A})\psi|^2 + a|\psi|^2 + \frac{b}{2} |\psi(\mathbf{r})|^4 \quad (2.65)$$

and the total free energy of the system becomes

$$F_s(T) = F_n(T) + \int \left(\frac{1}{2m^*} |(-i\hbar\nabla + 2e\mathbf{A})\psi|^2 + a|\psi|^2 + \frac{b}{2} |\psi|^4 \right) d^3\mathbf{r} \quad (2.66)$$

again, by imposing $\delta F = 0$ for any $\delta\psi$ we get the effective Schrödinger equation

$$\frac{1}{2m^*} (-i\hbar\nabla + 2e\mathbf{A})^2\psi + a\psi + b\psi|\psi|^2 = 0 \quad (2.67)$$

Now we can show that the Ginzburg-Landau model is a good theory of superconductivity by showing that it successfully predicts the London equation. To do this we first have to prove the following

$$\mathbf{J} = -\frac{\partial F}{\partial \mathbf{A}} \quad (2.68)$$

where is the functional derivative, which is defined by the relation

$$\delta F = \int \frac{\partial F}{\partial \mathbf{A}} \cdot \delta \mathbf{A} d^3\mathbf{r} \quad (2.69)$$

we start by evaluating the work done by the electromagnetic field on the system, which is equal to the change in the free energy,

$$\delta F = \delta W = dt \int \mathbf{J} \cdot \mathbf{E} d^3\mathbf{r} \quad (2.70)$$

In stationary conditions equation 2.4 takes the form $\nabla \times \mathbf{B} = \mu_0 \mathbf{J}$, thus we can rewrite δF as

$$\delta F = dt \int (\nabla \times \mathbf{B}) \cdot \mathbf{E} d^3 \mathbf{r} \quad (2.71)$$

because

$$\nabla \cdot (\mathbf{a} \times \mathbf{b}) = (\nabla \times \mathbf{a}) \cdot \mathbf{b} - \mathbf{a} \cdot (\nabla \times \mathbf{b}) \quad (2.72)$$

we obtain

$$\delta F = -dt \int \nabla \cdot (\mathbf{E} \times \mathbf{B}) d^3 \mathbf{r} + dt \int \mathbf{B} \cdot (\nabla \times \mathbf{E}) d^3 \mathbf{r} \quad (2.73)$$

the first integrale can be rewritten as an integral over an infinite surface using the divergence theorem and, since the fields vanish at an infinite distance from their sources, we have

$$\int \nabla \cdot (\mathbf{E} \times \mathbf{B}) d^3 \mathbf{r} = \oint (\mathbf{E} \times \mathbf{B}) \cdot \hat{n} d\sigma = 0 \quad (2.74)$$

using Maxwell's equation 2.2 we obtain

$$\begin{aligned} \delta F &= -dt \int \mathbf{B} \cdot \frac{\partial \mathbf{B}}{\partial t} d^3 \mathbf{r} = - \int \mathbf{B} \cdot \delta \mathbf{B} d^3 \mathbf{r} \\ &= - \int \mathbf{B} \cdot \delta (\nabla \times \mathbf{A}) d^3 \mathbf{r} = - \int \mathbf{B} \cdot (\nabla \times \delta \mathbf{A}) d^3 \mathbf{r} \end{aligned} \quad (2.75)$$

again, using equation 2.72, this expression can be rewritten as

$$\delta F = -\frac{1}{\mu_0} \int (\nabla \times \mathbf{B}) \cdot \delta \mathbf{A} d^3 \mathbf{r} = - \int \mathbf{J} \cdot \delta \mathbf{A} d^3 \mathbf{r} \quad (2.76)$$

finally, from 2.69 we get 2.68.

Now

$$\begin{aligned} \delta F &= \delta \int \frac{1}{2m^*} |(-i\hbar \nabla + 2e\mathbf{A})\psi|^2 d^3 \mathbf{r} \\ &= \frac{1}{2m^*} \int \delta((-i\hbar \nabla \psi + 2e\mathbf{A}\psi) \cdot (i\hbar \nabla \psi^* + 2e\mathbf{A}\psi^*)) d^3 \mathbf{r} \\ &= \frac{1}{2m^*} \int (2e\psi \delta \mathbf{A} \cdot (i\hbar \nabla \psi^* + 2e\mathbf{A}\psi^*) + 2e\psi^* \delta \mathbf{A} \cdot (-i\hbar \nabla \psi + 2e\mathbf{A}\psi)) d^3 \mathbf{r} \\ &= \frac{e}{m^*} \int (i\hbar \psi \nabla \psi^* - i\hbar \psi^* \nabla \psi + 4e|\psi|^2 \mathbf{A}) \cdot \delta \mathbf{A} d^3 \mathbf{r} \end{aligned} \quad (2.77)$$

Now, from 2.68 we get

$$\mathbf{J} = -\frac{\partial F}{\partial \mathbf{A}} = -\frac{e\hbar i}{m^*}(\psi^*\nabla\psi - \psi\nabla\psi^*) - \frac{4e^2}{m^*}|\psi|^2\mathbf{A} \quad (2.78)$$

in the bulk of the material the order parameter can be considered uniform, therefore $\nabla\psi = \nabla\psi^* = \mathbf{0}$ and we can write

$$\mathbf{J} = -\frac{4e^2|\psi|^2}{m^*}\mathbf{A} \quad (2.79)$$

which is the London equation. Comparing with and we get

$$\frac{n_se^2}{m} = \frac{4e^2|\psi|^2}{m^*} = \frac{1}{\mu_0\lambda^2} \quad (2.80)$$

By taking the value $m^* = \frac{m}{2}$ for the effective mass we get

$$|\psi|^2 = \frac{n_s}{2} \quad (2.81)$$

which gives us a hint for the physical meaning of the order parameter ψ and the effective mass m^* : ψ is the coherent wave function of the Cooper pairs and m^* is the mass of a single Cooper pair. Equation combined with gives an expression for the penetration depth

$$\lambda(T) = \sqrt{\frac{mb}{2\mu_0 e^2 \dot{a}(T - T_c)}} \quad (2.82)$$

Clearly this will diverge at the critical temperature. Since both λ and ξ are proportional to $(T - T_c)^{-\frac{1}{2}}$, the dimensionless ratio

$$k = \frac{\lambda(T)}{\xi(T)} \quad (2.83)$$

is independent of temperature within the Ginzburg-Landau theory. Table summarizes the measured values of penetration depth and coherence length at zero temperature for a selection of superconductors.

Superconductor	T_c (K)	$\lambda(0)$ (nm)	$\xi(0)$ (nm)	k
Al	1.18	1150	45	0.03
Sn	3.72	180	42	0.23
Pb	7.20	87	39	0.48
Nb	9.25	39	52	1.3
Nb ₃ Ge	23.2	3	90	30
YNi ₂ B ₂ C	15	8.1	103	12.7
K ₃ C ₆₀	19.4	2.8	240	95
YBa ₂ Cu ₃ O _{7-δ}	91	1.65	156	95

Table 2.1: Penetration depth, $\lambda(0)$, and coherence length, $\xi(0)$, at zero temperature for some important superconductors.

2.2.6 The BCS theory

2.3 Proximity Effects

2.3.1 Shift in the resonant field

Modified Kittel equation

$$\omega_{res} = \gamma \sqrt{(H_{res} + H_a)(H_{res} + H_a + M_{eff})} \quad (2.84)$$

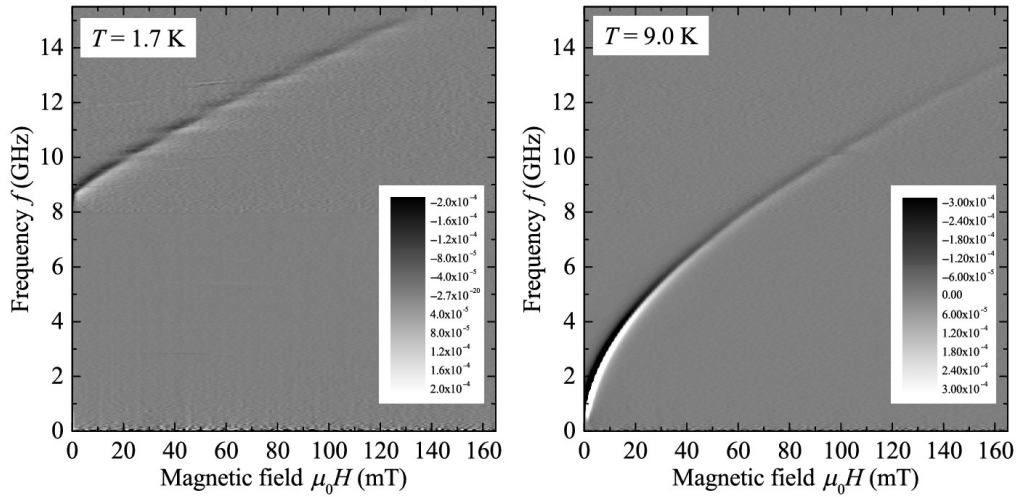


Figure 2.12: FMR absorption spectra for a sample above and below the critical temperature. The grayscale is coded in absolute units.

Power law

$$H_a(T) = H_0 \left(1 - \left(\frac{T}{T_c} \right)^p \right) \quad (2.85)$$

Chapter 3

Experimental Methods and Setup

In this investigation, a total of five experiments were conducted. The first two experiments aimed to reproduce the results reported in [1] by demonstrating that the shift in the resonant field follows the behavior described in the paper: it appears only below a certain critical temperature (indicating its connection to superconductivity) and increases as the temperature decreases. These experiments also aimed to show that both superconducting layers are required for this effect to occur. This was verified by performing measurements on a sample containing only one superconducting layer. The remaining three experiments were designed to test the dynamic effect theory presented in [6] by examining how the anisotropy field H_a changes under different conditions: introducing an additional metallic layer between the ferromagnetic and superconducting layers, varying the microwave power, and altering the thickness of the ferromagnetic layer.

3.1 Sample fabrication

The samples used in this study consisted of multilayer thin films with a layer of niobium (Nb), which is a type II superconductor with a critical temperature of 9 K, a ferromagnetic layer of permalloy $\text{Ni}_{80}\text{Fe}_{20}$ (Py), followed by another niobium layer. The films were deposited on silicon (Si) substrates, which were cut into $5\text{ mm} \times 5\text{ mm}$ squares and cleaned by sonication in acetone for 2 minutes, followed by sonication in isopropanol (IPA) for 2 minutes. This cleaning process is necessary to ensure a good adhesion of the samples to the substrate and a uniform thickness of the samples and to make the experiment reproducible. In one of the experiment two copper (Cu) layers

with variable thickness were introduced between both the superconducting and the ferromagnetic layers, figure 3.1 shows the structure of the samples used in this experiment.

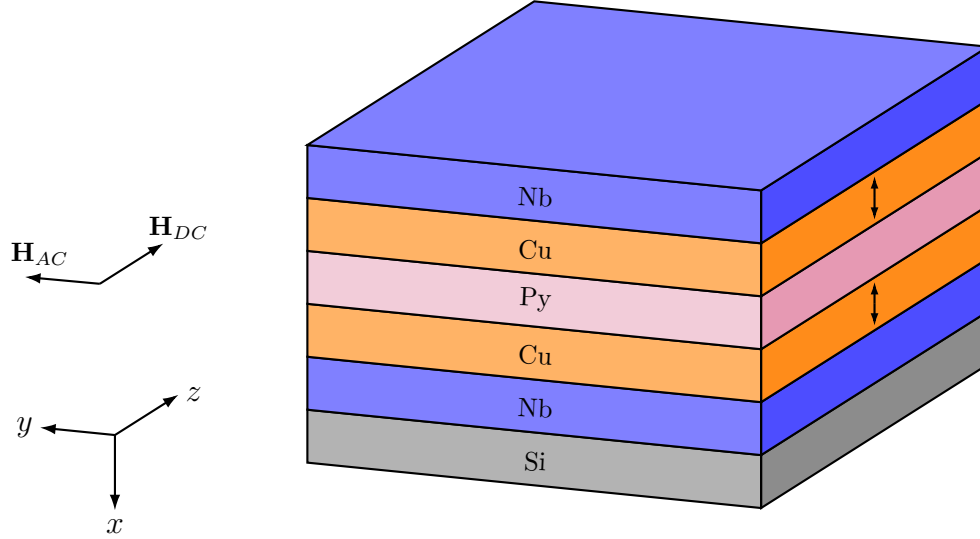


Figure 3.1: Scheme of the samples used in this study. The double arrows indicate the variable thickness of the copper layers. The figure also shows the orientation of the AC and DC magnetic fields used in the FMR measurements.

This samples were grown by DC magnetron sputtering, a physical vapour deposition (PVD) technique widely employed for the fabrication of metallic and multilayer thin films because it ensures high deposition rates and uniform coatings. Furthermore it allowed to grow the different layers in the same process, thus avoiding the exposure of intermediate layers and ensuring a good interface quality. In this method, an inert gas (argon in this case) is introduced into an ultra-high-vacuum chamber, and a high voltage is applied between the negatively charged target and the positively charged substrate. If the target material is conductive, then it is connected to direct current (DC) power supply and if the target material is insulator then it is connected to radio frequency (RF) power supply. Under the applied voltage, the gas is ionised, forming a plasma composed of positively charged ions and free electrons. The ions are accelerated by the electric field and strike the target with sufficient energy to eject atoms from it. These atoms then travel through the chamber and deposit as thin films on the substrate positioned in their path which is kept rotating to guarantee uniform sample thickness. In magnetron sputtering the plasma is confined near the target by a magnetic

field that forces electrons to follow a helical trajectory around the magnetic field lines. This increases both the sputtering yield and the overall deposition efficiency. The properties of the deposited films can be adjusted by changing sputtering parameters such as gas pressure, target–substrate distance and system temperature. Figure 3.2 illustrates the general scheme of a sputtering device.

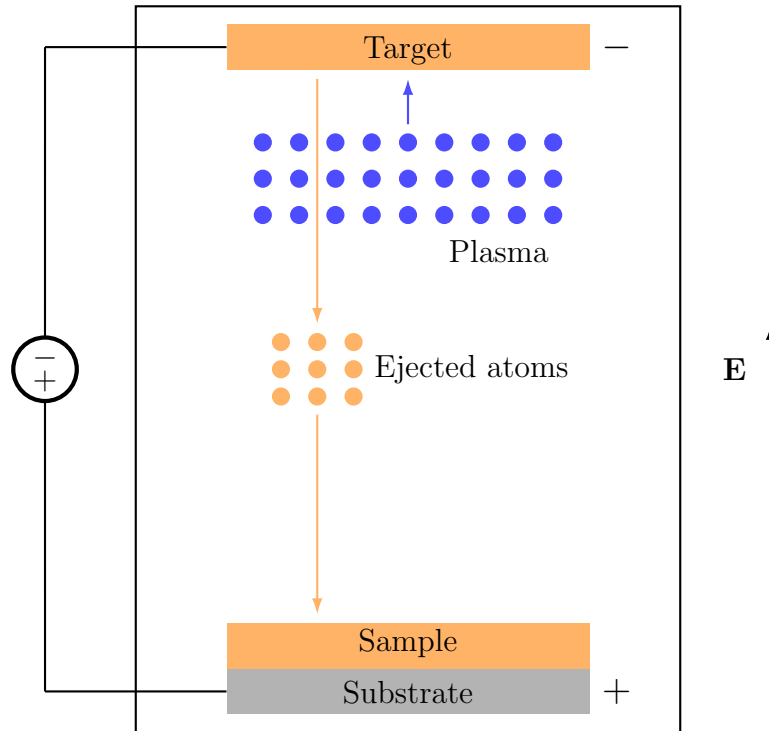


Figure 3.2: Schematic illustration of a sputtering device.

The structures investigated in this work are listed in the following table. Table 3.1 shows the thicknesses of the layers of the samples used in the control sample, bilayer and microwave power experiments (these experiments used a single sample). Table 3.2 and table 3.3 show the thicknesses of the layers for the copper thickness and permalloy thickness experiments respectively.

Sample	Nb thickness (nm)	Py thickness (nm)
Control Sample	80 ± 2.4	0
Bilayer	80 ± 2.4	2 ± 0.06
Microwave Power	80 ± 2.4	5 ± 0.15

Table 3.1: List of samples used in this study.

Sample	Nb thickness (nm)	Cu thickness (nm)	Py thickness (nm)
S ₀	80 ± 2.4	0	20 ± 0.6
S ₁	80 ± 2.4	2 ± 0.06	20 ± 0.6
S ₂	80 ± 2.4	5 ± 0.15	20 ± 0.6
S ₃	80 ± 2.4	10 ± 0.3	20 ± 0.6
S ₄	80 ± 2.4	20 ± 0.6	20 ± 0.6
S ₅	80 ± 2.4	50 ± 1.5	20 ± 0.6
S ₆	80 ± 2.4	100 ± 3	20 ± 0.6
S ₇	80 ± 2.4	200 ± 6	20 ± 0.6

Table 3.2: List of samples used in this study.

Sample	Nb thickness (nm)	Cu thickness (nm)	Py thickness (nm)
S ₀	80 ± 2.4	0	20 ± 0.6
S ₁	80 ± 2.4	2 ± 0.06	20 ± 0.6
S ₂	80 ± 2.4	5 ± 0.15	20 ± 0.6
S ₃	80 ± 2.4	10 ± 0.3	20 ± 0.6
S ₄	80 ± 2.4	20 ± 0.6	20 ± 0.6
S ₅	80 ± 2.4	50 ± 1.5	20 ± 0.6
S ₆	80 ± 2.4	100 ± 3	20 ± 0.6
S ₇	80 ± 2.4	200 ± 6	20 ± 0.6

Table 3.3: List of samples used in this study.

The samples were characterized using atomic force microscopy (AFM), which is a type of scanning probe microscopy that maps the surface topography of a sample with nanometer resolution by mechanically probing it with a sharp tip. The core of an AFM consists of a cantilever with a sharp tip that interacts with the sample surface through intermolecular forces between the atoms at the tip and the sample that cause the cantilever to deflect. This deflection is measured by a laser beam reflected from the cantilever onto a

position-sensitive photodiode. A feedback circuit maintains a constant interaction between the tip and the surface by adjusting the vertical position of the scanner. The height adjustments required to maintain the set interaction force are recorded to reconstruct the surface topography. This allowed to calculate the 3% errors on the thickness of the layers. Figure 3.3 shows the general scheme of a setup used for AFM measurements.

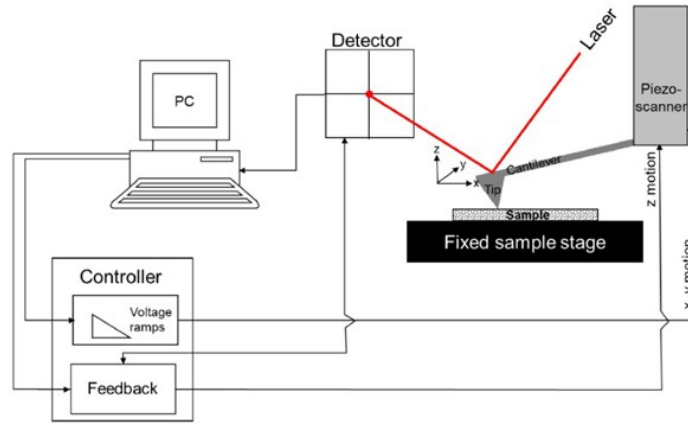


Figure 3.3: Schematic representation of an AFM setup.

3.2 Experimental Setup

The samples are mounted using vacuum grease onto a coplanar waveguide (CPW) and placed into a cryostat. A CPW is a planar transmission line structure consisting of a central conducting strip separated by narrow gaps from two ground planes, all defined on the same surface of a dielectric substrate. The signal propagates along the central conductor, while the two adjacent ground planes provide a return path. The sample is placed upside down (with the upper superconducting layer of figure 3.1 in contact with the CPW) because the magnetic field generated by the current in the CPW is stronger in the small region close to the surface and, due to the substrate being much thicker than the other layers, placing it the other way would make the magnetic field in the Py layer too weak to be measured. The external magnetic field and cryogenic environment are provided by the Oxford Instruments Teslatron cryostat, which contains a superconducting vector magnet with 6 T in the z-direction, 2 T in the x- and y-directions, and a base temperature of 1.5K. The sample is mounted on a sample probe, where the thickness lies in the x-direction, and is designed so that the sample is positioned between the superconducting coils to maximise the uniformity of the field. Steel

coaxial cables run down the length of the sample probe and act as microwave transmission lines to the CPW. Connected to the top of the cryostat, there are copper coaxial cables. The first copper cable acts as the input and is connected to a microwave generator, which drives the AC field that excites the sample. The second cable is connected to a microwave detector, which is directly connected to a lock-in amplifier. Figure 3.4 shows a schematic diagram of the cryostat and probe assembly while figure 3.5 shows an actual picture of the probe and sample.

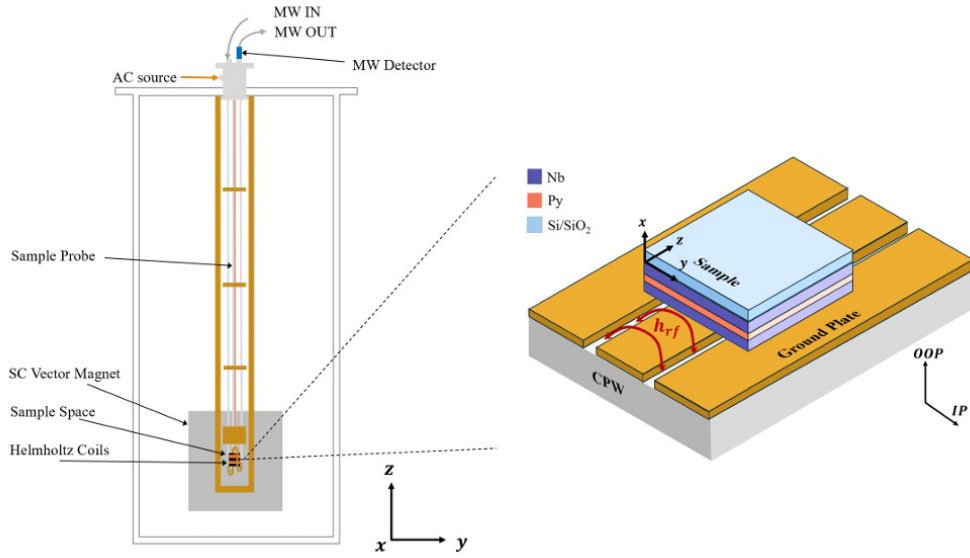


Figure 3.4: Schematic diagram of the cryostat and probe assembly.

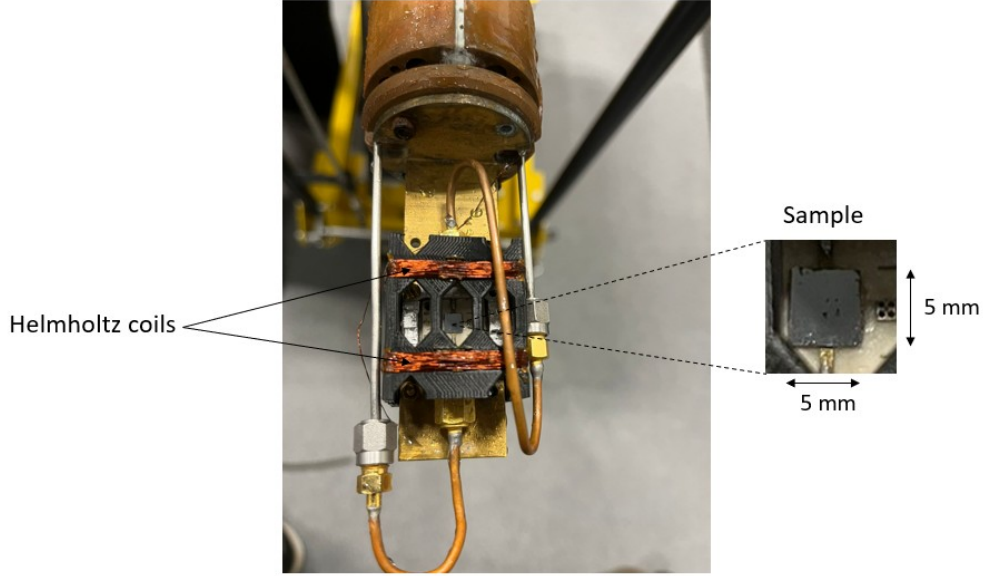


Figure 3.5: Picture of the sample and probe assembly.

Figure 3.6 shows a schematic of the complete setup used to conduct the FMR measurements. A signal is sent from a microwave source to the CPW through a coaxial cable, the current flowing in the CPW creates an oscillating magnetic field at the same microwave frequency as the input signal. According to Biot–Savart law, this field lies along the y axis and is approximately uniform across the sample. Together with the static magnetic field produced by the superconducting coil of the cryostat, which is constant in time, approximately uniform along the sample, and oriented along the z axis, this oscillating field can, in principle, induce ferromagnetic resonance. However, this effect is difficult to detect directly due to noise; therefore, a universal source, a pair of Helmholtz coils, and a lock-in amplifier are introduced to improve signal detection. Part of the energy carried by the signal is absorbed by the sample through FMR. The resulting transmitted signal is then sent to a diode, a composite component that converts the AC signal into a DC signal with an amplitude proportional to the average power of the input signal. The universal source sends a signal at 321 Hz to the two Helmholtz coils, generating a magnetic field whose frequency is much lower than the MW frequency of the input signal. This field can therefore be considered constant in time and added to the DC magnetic field generated by the electromagnet, the lock-in amplifier then registers the variation in the diode voltage δV_{diode} caused by a small variation δH_{DC} of the DC magnetic fields and returns a signal proportional to the ratio $\frac{\delta V_{diode}}{\delta H_{DC}}$ which is approximately equal to the

derivative of V_{diode} with respect to H_{DC} .

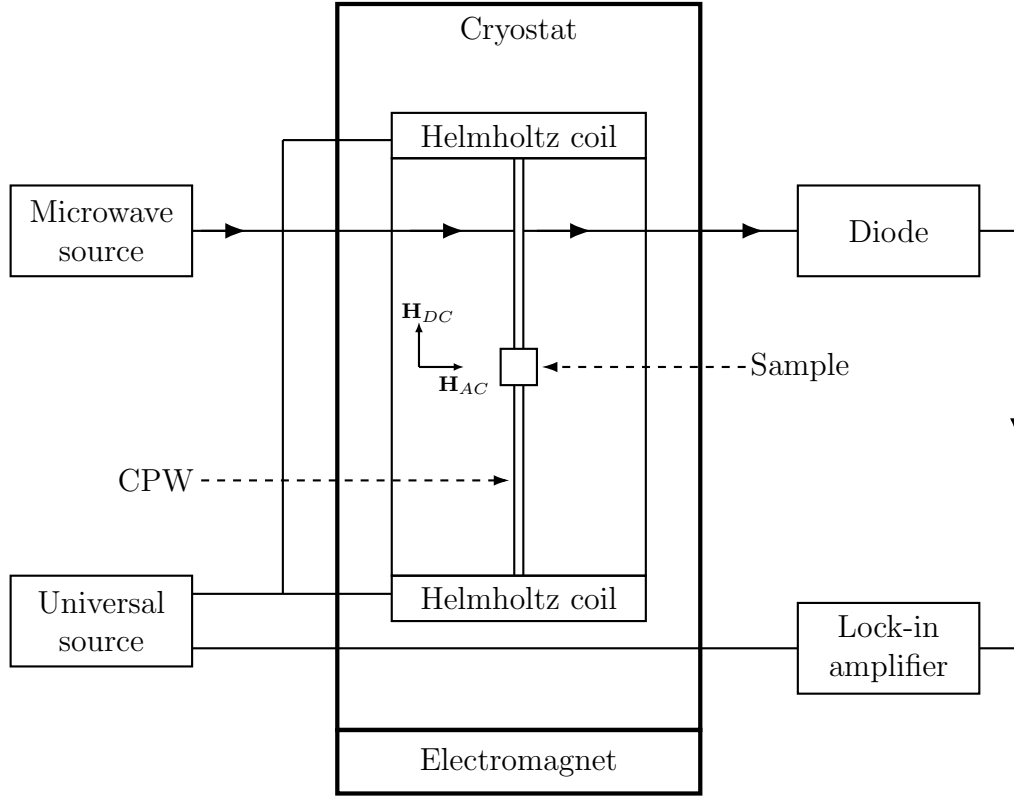


Figure 3.6: Schematic illustration of the setup used in the experiment.

3.3 FMR Measurements

Ferromagnetic resonance can be conducted by either sweeping frequency or magnetic field. In this case, the FMR absorption spectrum is detected by changing the external magnetic field at fixed frequencies to find the resonance field. As explained in the previous section, the resulting data is in the form of the derivative of the Lorentzian, which is analysed using the symmetric and antisymmetric lorentzian functions with respect to H_{res} . Another effect that is observed is that the data present a linear noise that is added to the lorentzian derivative function, therefore the profile of the data is expected to look like figure 3.7. The origin of the linear noise is still not known. A possible explanation for the linear noise is an impedance mismatch in the connections between the microwave generator and the CPW and between the CPW and the diode.

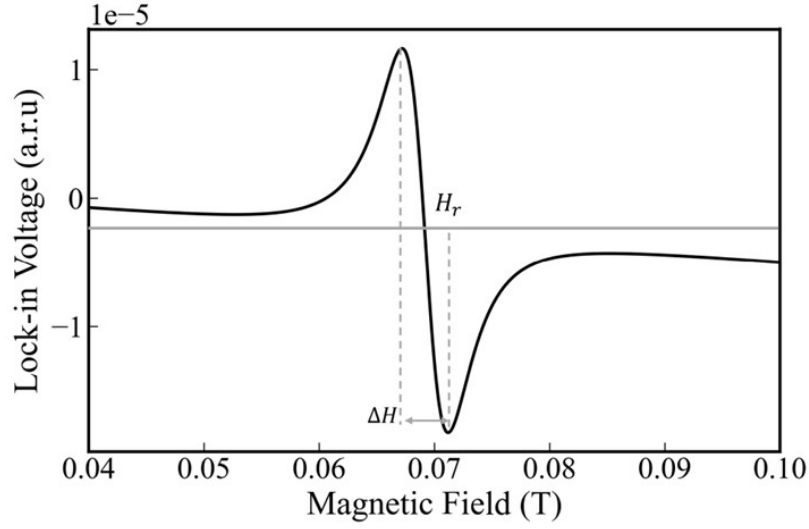


Figure 3.7: Expected behaviour of the signal

From this, the resonant field and linewidths can be extracted from each scan at each frequency and temperature by doing a fit on the data with a lorentzian derivative combined with a linear background. The values of resonant field and resonant frequency can be analyzed to form the Kittel relationship, allowing for exploration of changes in the resonant frequency-resonant field relationship. Apart from the bilayer sample, which is expected to follow the standard Kittel equation 2.27, the other samples are expected to exhibit anisotropy consistent with the modified Kittel equation 2.84.

Chapter 4

Data analysis and Results

4.1 Previous Literature (?)

The first step of the investigation was to reproduce the results reported in the literature [1]. Specifically, it is known that introducing two superconducting thin film layers around a ferromagnetic thin film generates a shift in the resonant field H_{res} , which therefore follows the modified Kittel equation 2.84 instead of the standard one 2.27. This shift appears only below a certain critical temperature, and both superconducting layers are required for the effect to occur. The first two conditions were verified using the control sample, while the last one was confirmed with the bilayer sample.

4.1.1 Control Sample

For different values of the temperature T and frequency f , the lock-in voltage was measured as a function of the applied DC magnetic field. All measurements were performed from 10 K down to 1.5 K over a frequency range of 6 to 15 GHz, using field-modulation FMR. As explained in the previous chapter, the signal detected by the lock-in amplifier is proportional to the derivative with respect to the magnetic field H of the signal from the CPW. Therefore, the data were fitted using the following equation:

$$V = S \frac{\Delta H^2 (H - H_{res})}{(\Delta H^2 + (H - H_{res})^2)^2} + A \frac{\Delta H (H - H_{res})^2 - \Delta H^3}{(\Delta H (H - H_{res})^2)^2} + aH + d \quad (4.1)$$

From this expression, the resonant field can be identified as the point where the function, excluding the linear background, crosses zero. Figure 4.1 shows a representative plot of the measured data.

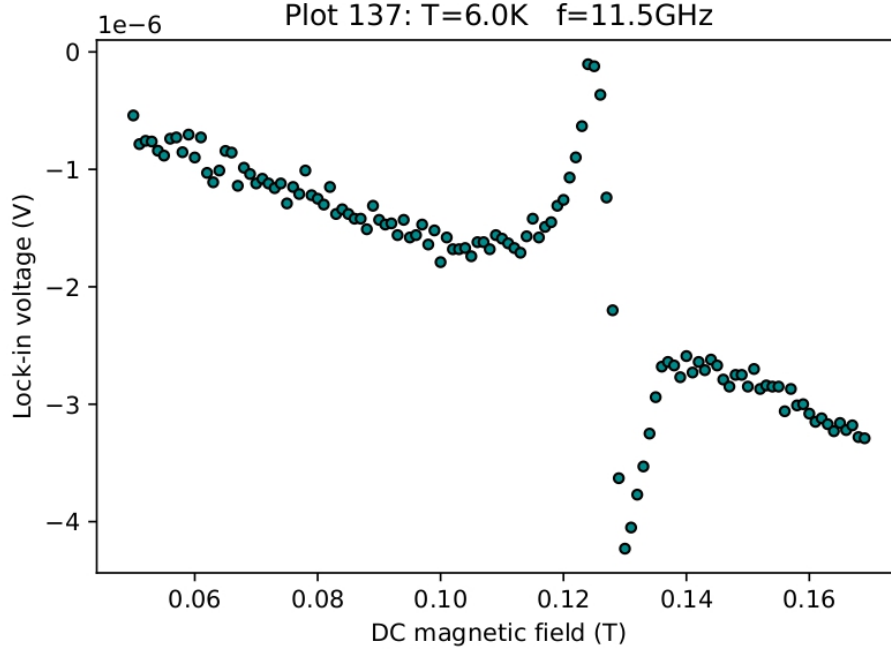


Figure 4.1: Experimental data for the DC magnetic field and lock-in voltage.

As expected, the plot follows the qualitative behaviour shown in Figure 3.7. To fit the data, the linear background was first removed. This was done by performing a linear fit over the entire dataset. Because the number of points around the resonant field was much smaller than the number of points in the linear regime, this approach provided a reliable estimate of the parameters a and d . Figure 4.2 shows the data from Figure 4.1 together with their fitted linear component.

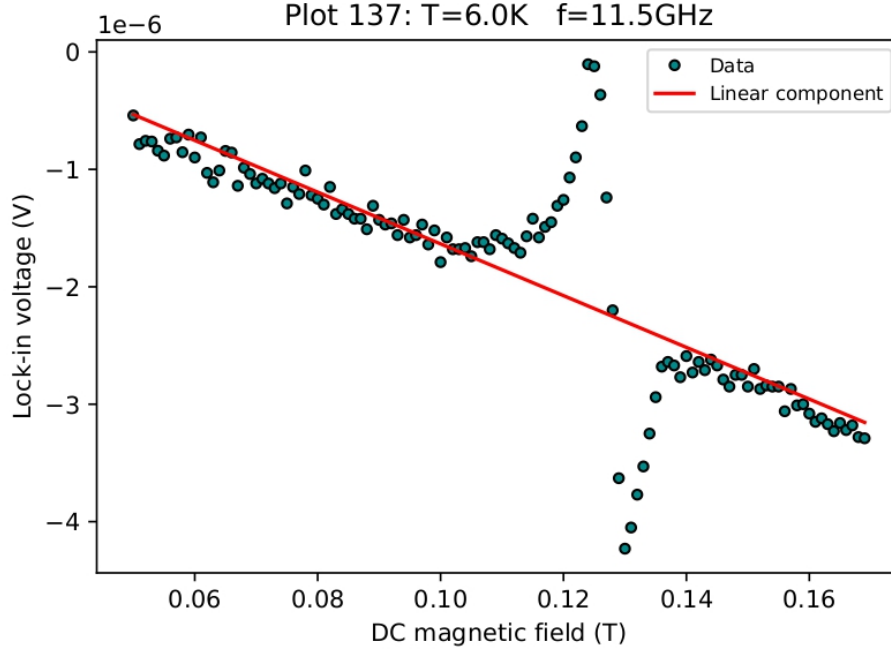


Figure 4.2: Linear component extrapolated from a fit on the entire dataset.

The linear component was then subtracted from the data and a four parameter fit was made on the resulting dataset. The initial guess values for the parameters H_{res} , ΔH , S , A were calculated using the following formulas:

$$H_{guess}^{res} = \frac{H_{max} + H_{min}}{2} \quad (4.2)$$

$$\Delta H_{guess} = 0.1(H_{max} - H_{min}) \quad (4.3)$$

$$S_{guess} = 0.7(V_{max} - V_{min}) \quad (4.4)$$

$$A_{guess} = 0.3(V_{max} - V_{min}) \quad (4.5)$$

Where V_{max} and V_{min} are the maximum and minimum value of the lock-in voltage in the data, while H_{max} and H_{min} are the corresponding values of the DC magnetic field. Figure 4.3 shows the same dataset together with the initial guess curve, figure 4.4 displays the final fitted curve.

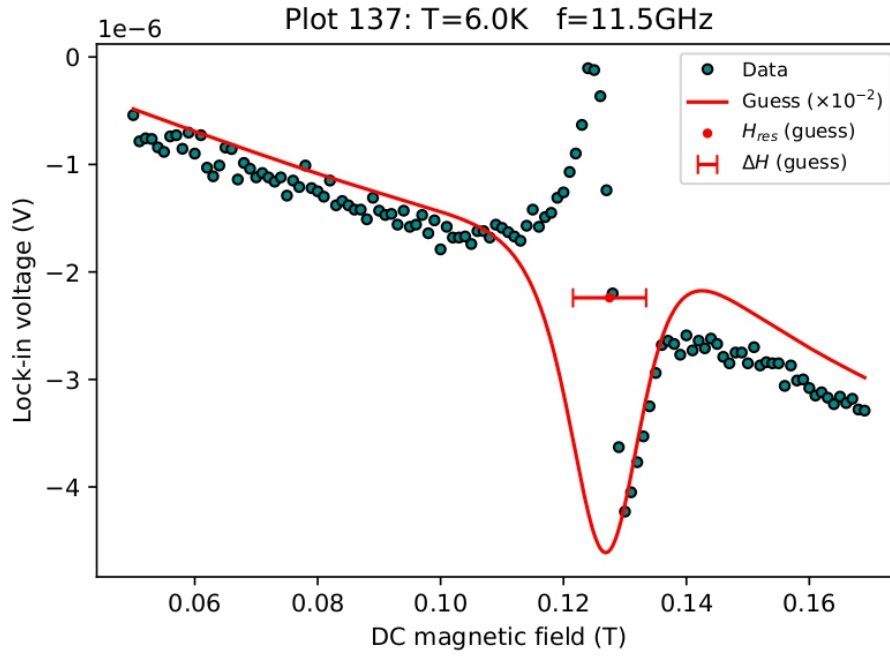


Figure 4.3: Guess curve obtained from equations 4.2 - 4.5.

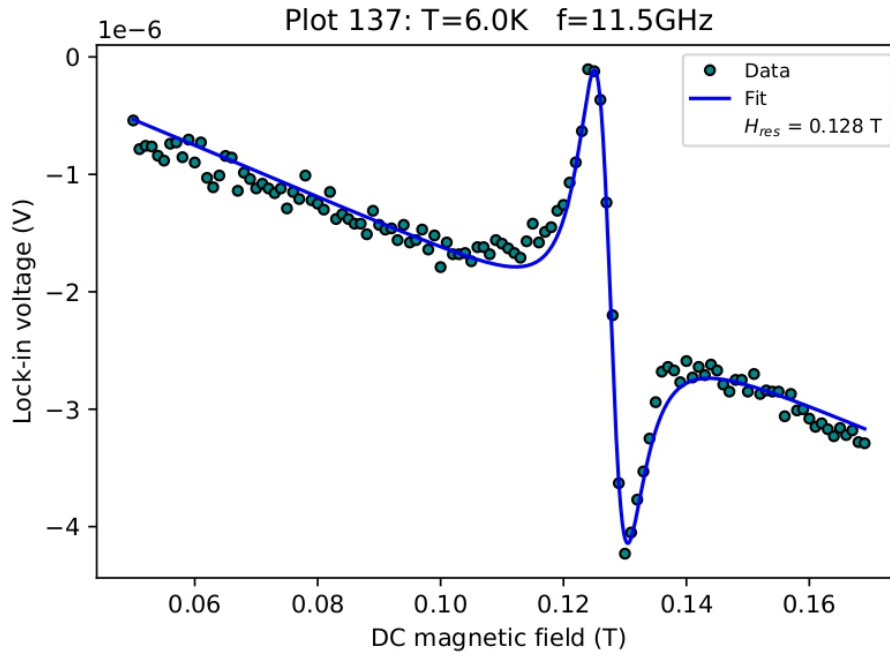


Figure 4.4: Fit curve obtained from the previous guess.

For each temperature the values of resonant frequency and field were fitted with the modified Kittel's equation

$$f_{res} = \frac{\gamma}{2\pi} \sqrt{(H_{res} + H_a)(H_{res} + H_a + M_{eff})} \quad (4.6)$$

to find the anisotropy field H_a . Figure 4.5 shows an example of a dataset together with its fitted curve.

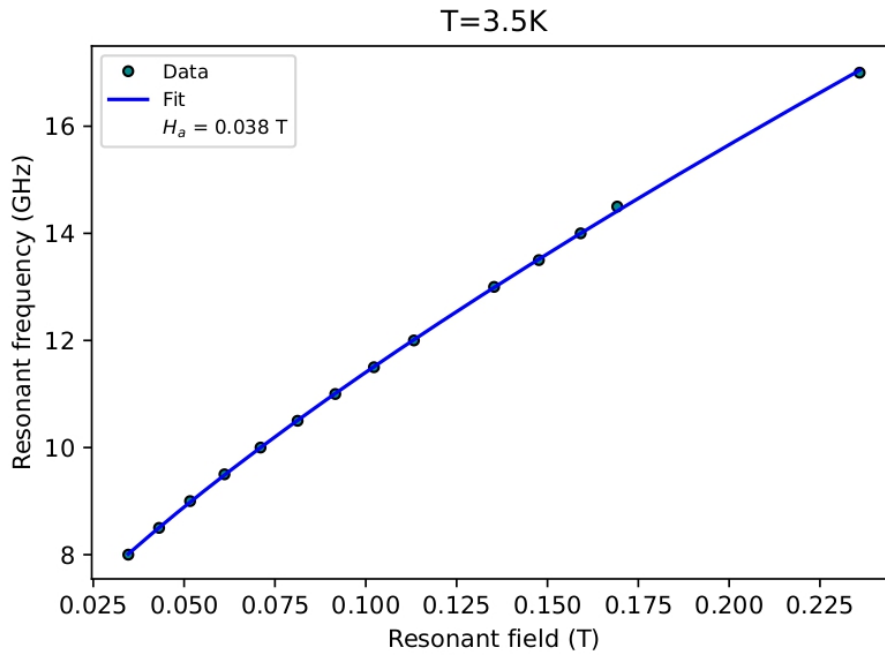


Figure 4.5: Experimental data for resonant field and resonant frequency at $T = 3.5K$.

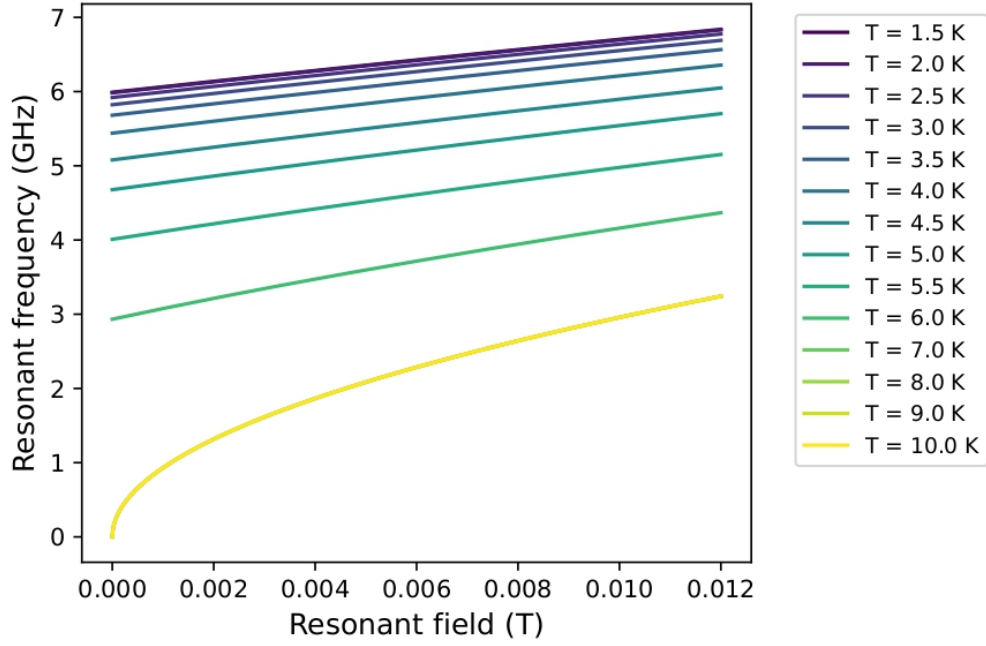


Figure 4.6: Fitted curves of the Kittel equation for different temperatures.

As expected, the curves below the critical temperature coincide, confirming that no change in the anisotropy occurs in this regime. In contrast, the curves above the critical temperature display an offset that increases as the temperature decreases. Finally, the values of the anisotropy field H_a were plotted as a function of the temperature T to verify that the shift in H_{res} appears only below the critical temperature T_c . The resulting plot is shown in Figure 4.7.

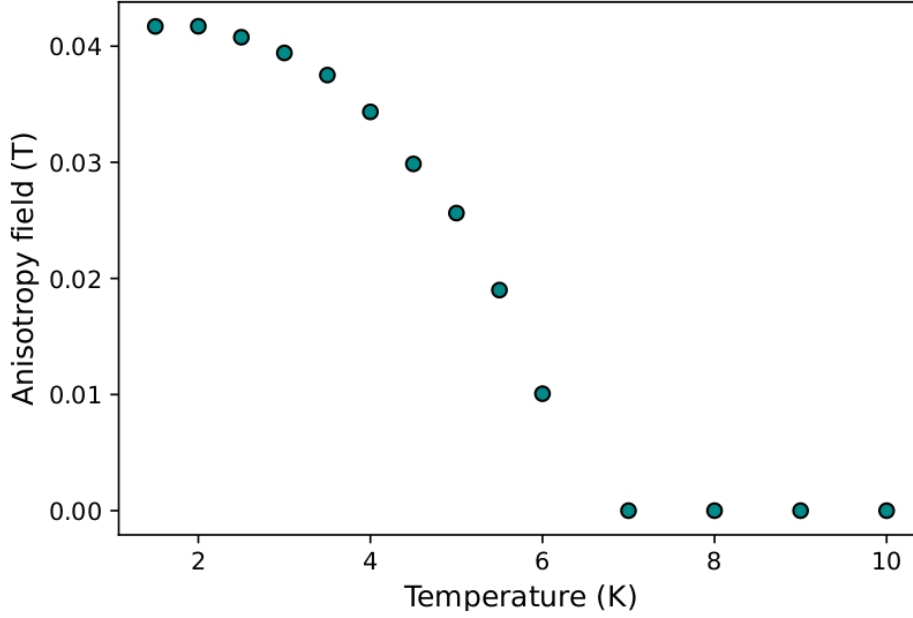


Figure 4.7: Anisotropy field as a function of temperature.

As expected, H_a has a constant value of zero below T_c . The observed critical temperature is lower than the theoretical critical temperature of niobium (9 K), this can be attributed to a proximity effect. This plot further confirms that superconductivity is a continuous phase transition, as no discontinuity is observed in H_a , and that the anisotropy field can be used as an order parameter for the trilayer system.

The data for the anisotropy field with $H_a > 0$ were then fitted with the power law 2.85 and the following values for the critical temperature T_C and for the exponent p were obtained

$$T_c = 6.5 \pm 0.027 \text{ K} \quad (4.7)$$

$$p = 3.489 \pm 0.095 \quad (4.8)$$

Figure 4.8 shows the comparison between the data and the fit curve

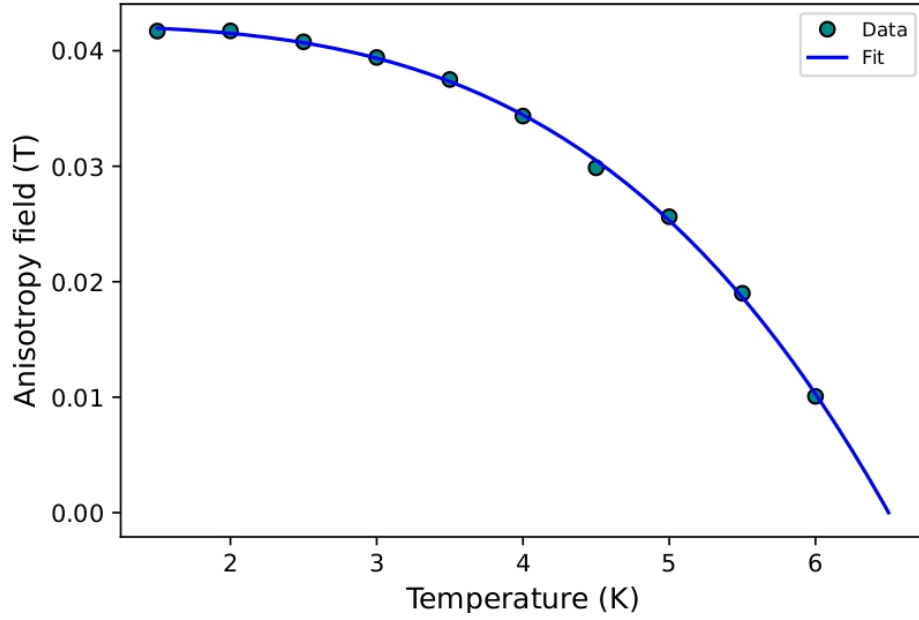


Figure 4.8: Values of the anisotropy field with the fit curve obtained from the power law below T_c .

4.1.2 Bilayer

The same procedure used for the control sample was used for the bilayer sample. Figure 4.9 shows the values of the anisotropy field for different temperatures both above and below the critical temperature.

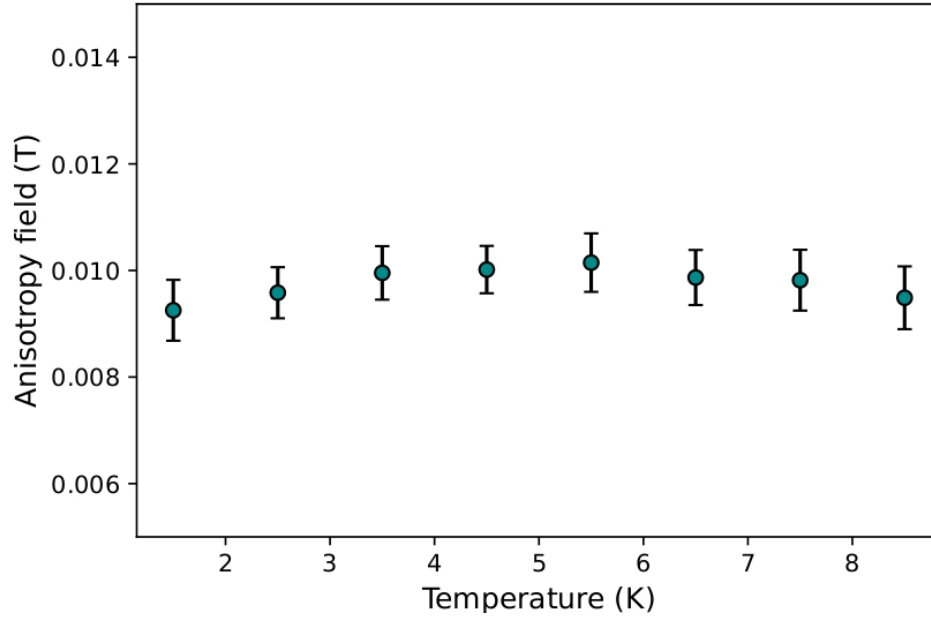


Figure 4.9: Anisotropy field for different temperatures in the asymmetric sample.

In this case there, no change in the anisotropy field is observed below the critical temperature within the experimental errors. This indicates that both superconducting layers are required in order to produce the change in anisotropy. Moreover, the anisotropy field remains greater than zero even below the critical temperature, this is due to shape anisotropy and impurities in the sample.

4.2 New Experiments (?)

4.2.1 Copper Thickness

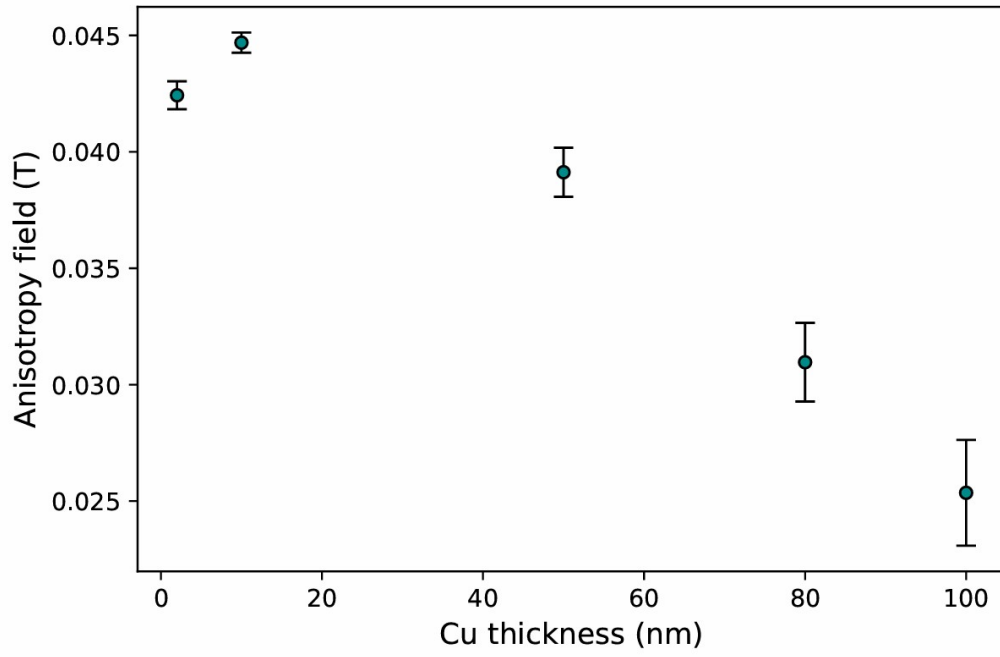


Figure 4.10: Anisotropy field at $T = 2.0K$ for different thicknesses of the Cu layers.

4.2.2 Microwave Power

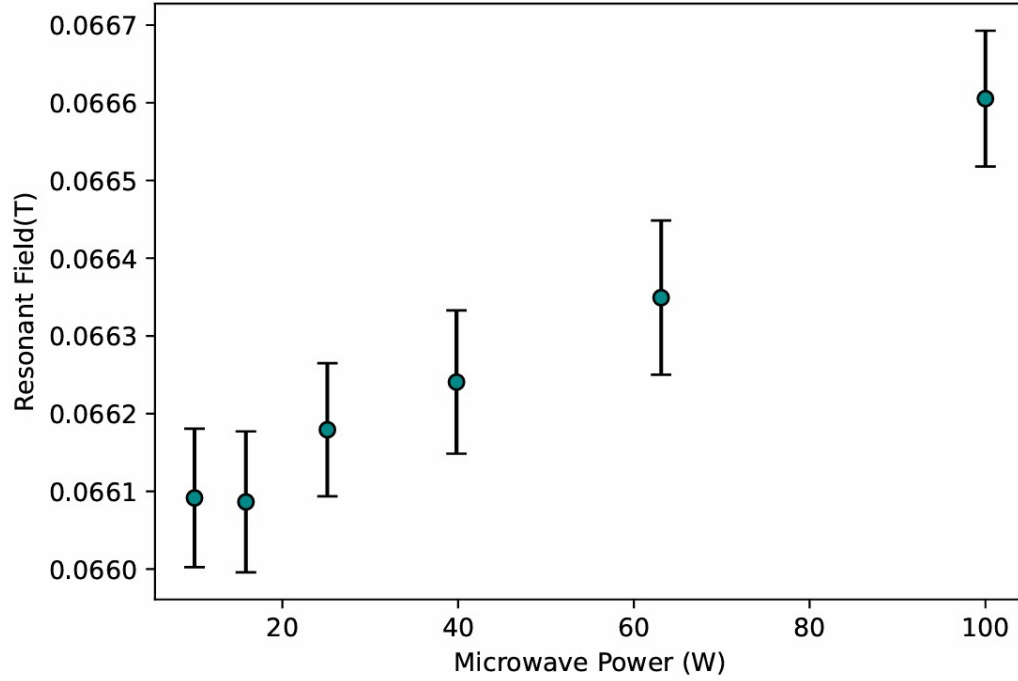


Figure 4.11: Anisotropy field at $T = 2.0K$ for different thicknesses of the Cu layers.

4.2.3 Permalloy Thickness

4.3 Results

Bibliography

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Aknowlesgements/Ringraziamenti